



Operation Guide

SIGMATRONIC
IR Industrial Battery Charger
Quick Start Procedure





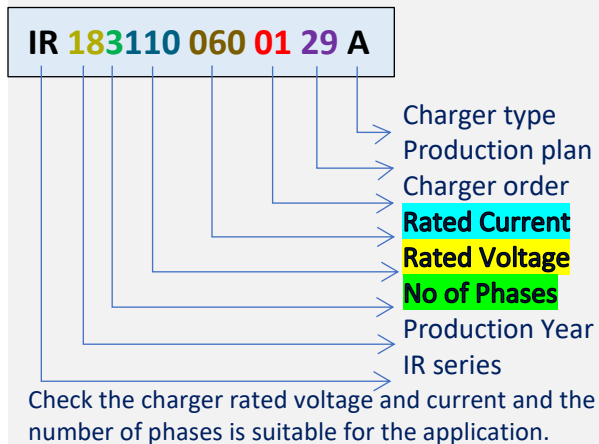
SIGMATRONIC IR Industrial Battery Charger Quick Start Procedure

OVERVIEW

The **IR series** provides the fully protection for DC critical industrial systems from unexpected side effects of power obstructions. It relies on microcontroller control to guarantee the needed flexibility and stability for different applications and loads. The following procedure is the wiring and operating of the charger.

Step 1: Model Identification

To verify the Charger unit is the correct one for the application you need, check its type, output ratings on the nameplate charger. This can be also identified from the serial number partitions as shown in this example.



Step 2: Prepare for Installation

The charger must be mounted in a vertical position only.
Avoid mounting close to high heat sources.
The mounting location must be free from frost.
Do not restrict the flow of air through the air filters or inlets and outlets of air within the charger.
Correct air clearance around the charger must be observed.
Cable from the back.
AC and DC cable connected to the charger should be sized properly according to the amperage.

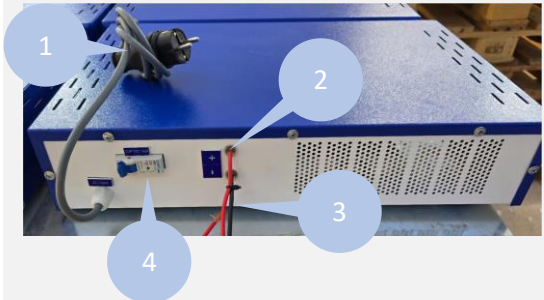
Step 3: Installation

Follow the standard installation procedure carefully before start up the battery charger. Ignoring the strict following to the instructions may lead to serious situations such personnel injury, property damage or equipment failure.
Make sure that main switch (within the charger or external) are in "off position".
Plug in the ac plug in the ac source.
Press the switch to ON position.
Check the display and the displayed voltage on it.
Switch off the charger then plugin the dc cables, check the polarization.
Switch on the charger and check the current on the display.
Use suitable tools to tie all the cables inside the terminals.

Note:

To prevent electric shock, make sure that the plug(socket) is connected to a reliable Earth.

Step 4: Connection Labeling



- 1- AC Plug.
- 2- DC "+" positive Output.
- 3- DC "-" negative Output.
- 4- Output Circuit Breaker.

Step 3: Installation



- 1- ON – OFF Button.
- 2- Display.

Sales & Head Office and Factory:

📍 Egypt, Cairo, El Obour City, Area No 22, 23, Block 13037, 1st Industrial zone.

www.sigma-engineer.com

✉ Info@sigma-engineer.com



SIGMATRONIC

IR Industrial Battery Charger

Quick Start Procedure

Step 4: Applying Power and Front panel



Mode	Function*
Normal	-The upper display displays the voltage value. -The lower display displays the current value. -The "ON-OFF" LED is ON. -The "Charge" LED is ON. -The "Limit" LED is OFF.
Current Limit1	-The upper display displays the voltage value. -The lower display swing between current value and "CrL1" -The "ON-OFF" LED is ON. -The "Charge" LED is ON. -The "Limit" LED is ON.
Current Limit2	-The upper display displays the voltage value. -The lower display swing between current value and "CrL2" -The "ON-OFF" LED is ON. -The "Charge" LED is ON. -The "Limit" LED is ON.
Overt Voltage	-The upper display displays "E-OV". -The lower display displays "----" -The "ON-OFF" LED is OFF. -The "Charge" LED is OFF. -The "Limit" LED is OFF.
Overt Load	-The upper display displays "----". -The lower display displays "E-OL" -The "ON-OFF" LED is OFF. -The "Charge" LED is OFF. -The "Limit" LED is OFF.
Short Circuit	-The upper display displays "----". -The lower display displays "E-OL" -The "ON-OFF" LED is OFF. -The "Charge" LED is OFF. -The "Limit" LED is OFF.

Models:

	IR-320-12	IR-320-24	IR-600-48
Wattage	290	290	580
Input voltage	100 ~ 240VAC	100 ~ 240VAC	100 ~ 240VAC
Output Current	20A	10A	10A
Output Voltage	Boost charge 14.2V, Float charge 13.6V	Boost charge 28.8V, Float charge 27.2V	Boost charge 57.6V, Float charge 54.4V
FREQUENCY RANGE	47 ~ 63Hz	47 ~ 63Hz	47 ~ 63Hz
Working Temperature	-30~70°C		
Safety Standards	UL62368-1, TUV BS EN/EN62368-1, EAC TP TC 004, .1, BSMI CNS15598-1, AS/NZS 60950.1, CCC GB4943 IS13252(Part1)/ IEC60950-1(except .5V,48V) Dekra EN 61558-1/2-16, IEC 61558-1/2-16		
EMC Immunity	Compliance to EN61000-4-2,3,4,5,6,8,11, EN55035, light industry level, BS EN/ BS EN/ EAC TP TC 020		

Features:

- 3-stage charging curve for charging mode.
- 30~70° wide operating temperature.
- Multiple protections: short circuit/ Over load/ Over voltage/ Over Temperature.
- Suitable for lead-acid (Pb) batteries.

Sales & Head Office and Factory:

📍 Egypt, Cairo, El Obour City, Area No 22, 23, Block 13037, 1st Industrial zone.

www.sigma-engineer.com

✉ Info@sigma-engineer.com